ISTANBUL UNIVERSITY-CERRAHPAŞA
Department of Computer Engineering
BIMU2059-DIFFERENTIAL EQUATIONS
Midterm (Weight, 50%)

Student's Name	
Student's ID Number	

Academic Year	2025-2026	**Date of Exam**	24.11.2025 (15:20)
Professor	Assoc. Prof. Dr. Uğur Odabaşı	**Time Duration (min)**	**65**

Total /100

QUESTIONS AND SOLUTIONS

1.

$$(\sqrt{x^2 - y^2} + y)\,dx - x\,dy = 0$$

a) Find the general solution.

b) Find any particular solution not obtainable from the general solution (singular solution).

$(15 + 10 = 25$ Pts)

Solution: First, observe that the term $\sqrt{x^2 - y^2}$ implies $|y| \leq |x|$. Since the coefficients are homogeneous, we use the substitution $y = ux$. Then $dy = u\,dx + x\,du$. Substituting these into the equation:

$$(\sqrt{x^2(1 - u^2)} + ux)\,dx - x(u\,dx + x\,du) = 0$$
$$(|x|\sqrt{1 - u^2} + ux)\,dx - (ux\,dx + x^2\,du) = 0$$

Assuming $x \neq 0$, we simplify. (Considering $x > 0$, so $|x| = x$):

$$\sqrt{1 - u^2}\,dx - x\,du = 0$$

To separate variables, we divide by $x\sqrt{1 - u^2}$. This requires $|u| < 1$ (i.e., $y \neq \pm x$).

Separating variables and integrating:

$$\frac{dx}{x} = \frac{du}{\sqrt{1 - u^2}} \implies \int \frac{dx}{x} = \int \frac{du}{\sqrt{1 - u^2}} \implies \ln|x| = \arcsin(u) + C$$

Substituting $u = y/x$ back:

$$\ln|x| - C = \arcsin\left(\frac{y}{x}\right) \implies \frac{y}{x} = \sin(\ln|x| - C)$$

General Solution:

$$\boxed{y = x\sin(\ln|x| + C_1)} \quad \text{valid for } |y| < |x|$$

Singular Solutions: During separation, we excluded $|u| = 1$. Checking these excluded functions $u = 1 \implies y = x$ and $u = -1 \implies y = -x$: By substitution, they verify the original equation. However, they are **not obtainable** from the general solution family (since sin cannot be constantly ± 1).

$$\boxed{y = x \quad \text{and} \quad y = -x}$$

2. Solve the initial value problem:

$$\frac{dy}{dx} = \frac{xy^2 - \cos x \sin x}{y(1-x^2)}, \quad y(0) = 2$$

(25 Pts)

Solution: Rewrite the equation in differential form:

$$(\cos x \sin x - xy^2)\,dx + y(1-x^2)\,dy = 0$$

Check for exactness with $M = \cos x \sin x - xy^2$ and $N = y(1-x^2)$:

$$\frac{\partial M}{\partial y} = -2xy \quad \text{and} \quad \frac{\partial N}{\partial x} = -2xy$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is exact.

Find the potential function $f(x,y)$ by integrating N with respect to y:

$$f(x,y) = \int y(1-x^2)\,dy = \frac{y^2}{2}(1-x^2) + h(x)$$

Differentiate $f(x,y)$ with respect to x and equate to M to find $h(x)$:

$$\frac{\partial f}{\partial x} = -xy^2 + h'(x) = \cos x \sin x - xy^2$$
$$h'(x) = \cos x \sin x$$
$$h(x) = \int \cos x \sin x\,dx = -\frac{1}{2}\cos^2 x$$

The general solution is $f(x,y) = c_1$:

$$\frac{y^2}{2}(1-x^2) - \frac{1}{2}\cos^2 x = c_1 \implies y^2(1-x^2) - \cos^2 x = C$$

Apply the initial condition $y(0) = 2$:

$$(2)^2(1-0^2) - \cos^2(0) = C \implies 4(1) - 1 = C \implies C = 3$$

Final Solution:

$$\boxed{y^2(1-x^2) - \cos^2 x = 3}$$

3. a) Prove that every separable first-order differential equation is exact.

   b) Given first order nonlinear differential equation

$$y' + \tfrac{7y}{x} - 3y^2 = \tfrac{3}{x^2}$$

   i) Prove that after substitution, $y(x) = \frac{1}{x} + u(x)$, the given equation can be transformed into a Bernoulli differential equation;

   ii) Solve the resulting Bernoulli differential equation.

(10 + 2 × 7.5 = 25 Pts)

Solution: The separable equation $\frac{dy}{dx} = g(x)h(y)$ is written in the differential form $M\,dx + N\,dy = 0$ as:

$$g(x)\,dx + \left(-\frac{1}{h(y)}\right)\,dy = 0$$

Let $M = g(x)$ and $N = -\frac{1}{h(y)}$. The exactness condition requires $M_y = N_x$:

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(g(x)) = 0 \text{ and } \frac{\partial N}{\partial x} = \frac{\partial}{\partial x}\left(-\frac{1}{h(y)}\right) = 0$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 0$, the equation is always exact.

b) Using the substitution $y = x^{-1} + u$ (and $y' = -x^{-2} + u'$), the original equation transforms:

$$\begin{aligned}
\left(-x^{-2} + u'\right) + \frac{7}{x}\left(x^{-1} + u\right) - 3\left(x^{-1} + u\right)^2 &= 3x^{-2} \\
u' + \left(-x^{-2} + 7x^{-2} - 3x^{-2}\right) + \left(\frac{7u}{x} - \frac{6u}{x}\right) - 3u^2 &= 3x^{-2} \\
u' + 3x^{-2} + \frac{u}{x} - 3u^2 = 3x^{-2} &\implies u' + \frac{1}{x}u = 3u^2
\end{aligned}$$

The resulting Bernoulli equation ($n = 2$) is solved using the substitution $w = u^{-1}$. This yields the linear ODE $w' - \frac{1}{x}w = -3$. The integrating factor is $\mu(x) = e^{\int -1/x dx} = 1/x$. Multiplying and integrating gives:

$$\frac{d}{dx}\left(\frac{1}{x}w\right) = -\frac{3}{x} \implies \frac{1}{x}w = -3\ln|x| + C$$

Solving for w: $w = x(C - 3\ln|x|)$. Back-substituting $u = w^{-1}$ and $y = x^{-1} + u$:

$$\boxed{y(x) = \frac{1}{x} + \frac{1}{x(C - 3\ln|x|)}}$$

4. Verify that the functions $\{1, x, \cos x, \sin x\}$ form a fundamental set of solutions of the differential equation

$$y^{(4)} + y'' = 0$$

on the interval $(-\infty, \infty)$, and form the general solution.

(25 Pts)

Solution:

1. Verification of Solutions

The differential equation is $y^{(4)} + y'' = 0$.

- For $y_1 = 1$: $y_1^{(4)} + y_1'' = 0 + 0 = 0$.
- For $y_2 = x$: $y_2^{(4)} + y_2'' = 0 + 0 = 0$.
- For $y_3 = \cos x$: $y_3^{(4)} + y_3'' = (\cos x) + (-\cos x) = 0$.
- For $y_4 = \sin x$: $y_4^{(4)} + y_4'' = (\sin x) + (-\sin x) = 0$.

All four functions are solutions to the fourth-order homogeneous equation.

2. Verification of Linear Independence (Wronskian)

We calculate the Wronskian $W(y_1, y_2, y_3, y_4)$:

$$W(x) = \begin{vmatrix} 1 & x & \cos x & \sin x \\ 0 & 1 & -\sin x & \cos x \\ 0 & 0 & -\cos x & -\sin x \\ 0 & 0 & \sin x & -\cos x \end{vmatrix}$$

Expanding the determinant along the first column (and then the first column again):

$$\begin{aligned} W(x) &= 1 \cdot 1 \cdot \begin{vmatrix} -\cos x & -\sin x \\ \sin x & -\cos x \end{vmatrix} \\ W(x) &= (-\cos x)(-\cos x) - (-\sin x)(\sin x) \\ W(x) &= \cos^2 x + \sin^2 x = 1 \end{aligned}$$

Since $W(x) = 1 \neq 0$ for all $x \in (-\infty, \infty)$, the solutions are **linearly independent** and form a Fundamental Set of Solutions.

3. General Solution

The general solution is the linear combination of the Fundamental Set:

$$y(x) = c_1 y_1 + c_2 y_2 + c_3 y_3 + c_4 y_4$$

$$\boxed{y(x) = c_1 + c_2 x + c_3 \cos x + c_4 \sin x}$$